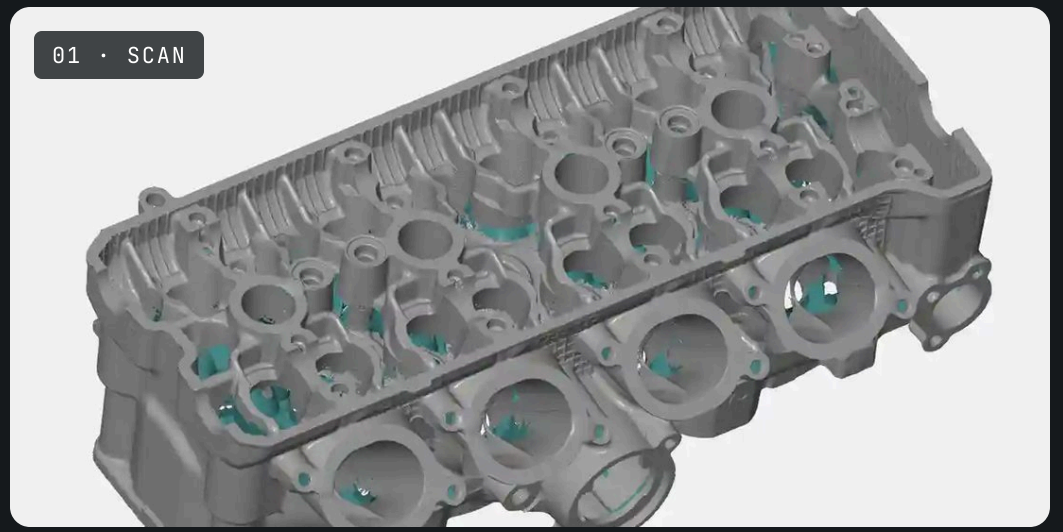
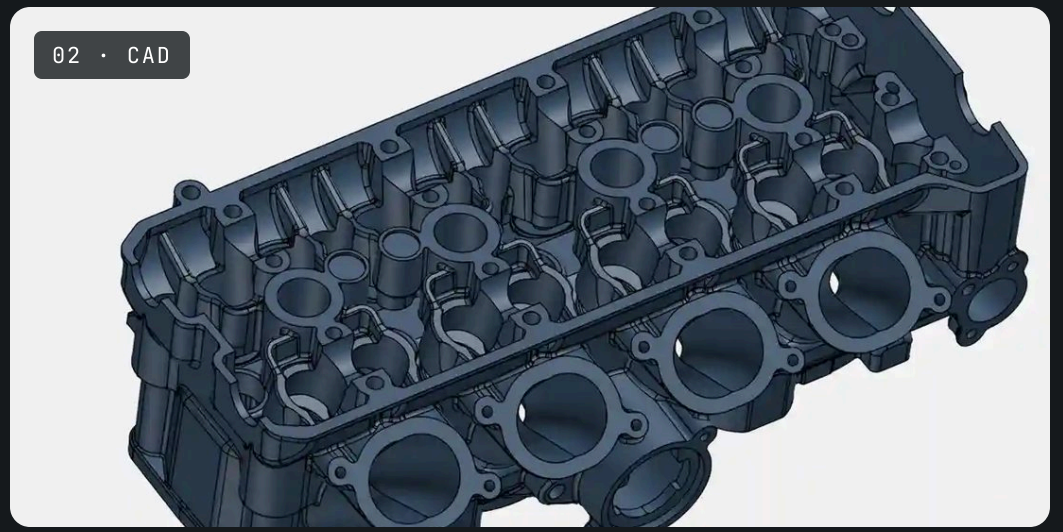
VSHARK
ENGINEERING SOLUTIONS

3D SCANNING & 3D DESIGN SERVICES

We capture reality. Then we engineer it.

Precise 3D scanning and innovative 3D design that transform physical parts, products and spaces into digital models — and back into physical prototypes.





EMAIL
info@vsharksolutions.com

INDIA
+91 95142 48525

UAE
+971 52 248 6238

(01) — WHO WE ARE

From physical to digital and back.

VShark Solutions provides state-of-the-art 3D scanning and 3D design services for businesses and individuals.

We capture parts, products and spaces as accurate digital data — then engineer that data into editable CAD, flawless surfaces, BIM models, inspection reports, prototypes and photoreal visuals. One partner, from the physical object to the finished digital deliverable and back again.

0.02 mm SCAN ACCURACY	7 CORE SERVICES
6 INDUSTRIES SERVED	6+ COUNTRIES WITH REPEAT CLIENTS

Seven services, one partner.

Use one service on its own, or hand us the whole job — from first scan to final file.



01

3D Scanning

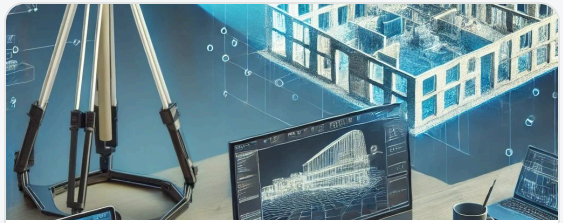
Capture any part, product or space as an accurate digital model.



02

Reverse Engineering

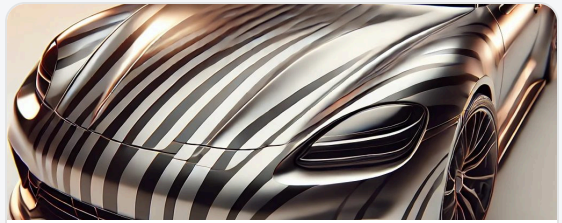
Turn scan data into fully editable, manufacturable CAD.



03

Scan to BIM

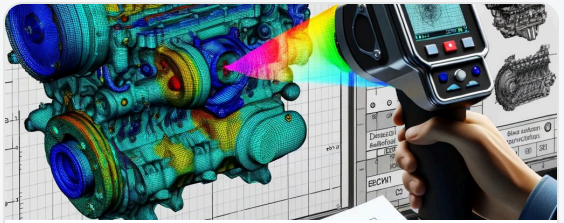
Existing buildings captured as accurate, intelligent BIM models.



04

Class A Surfacing

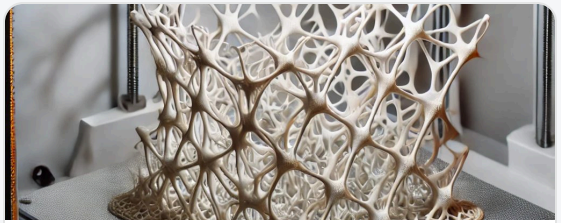
Flawless, curvature-continuous surfaces for visible parts.



05

Digital Metrology

Compare manufactured parts to their CAD — to the micron.



06

3D Printing

Rapid prototypes and functional parts, straight from your model.



07

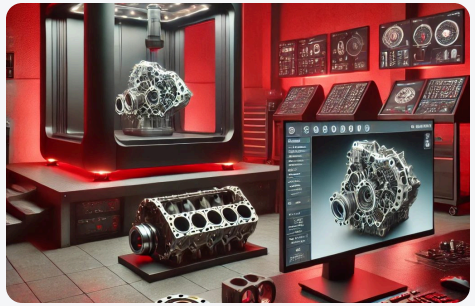
Product Rendering

Photoreal visuals that sell a product before it is ever made.

End-to-end delivery

Scan, engineer, inspect, print and render — under one roof, with one point of contact.

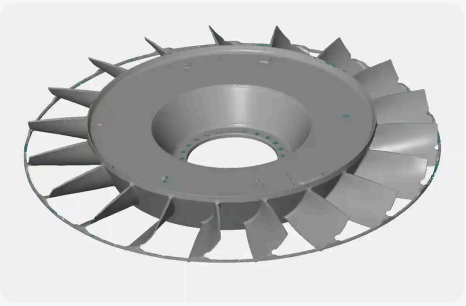
A simple, four-step process — scan to solution.



01

Capture

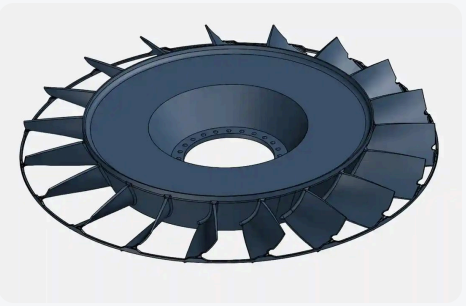
We scan the part, product or site — in our lab or on location.



02

Process

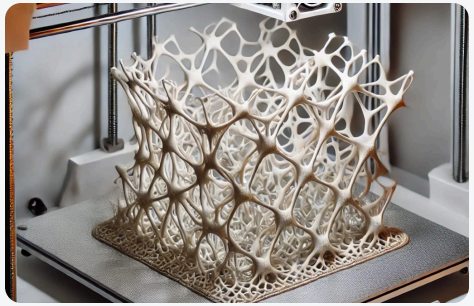
Scans are aligned, cleaned and merged into usable, watertight data.



03

Engineer

CAD, Class A surfaces, BIM or inspection built from that data.



04

Deliver

Native files, drawings and reports in the formats your team uses.



(02.1) — SERVICES

3D Scanning

Capture any part, product or space as an accurate digital model.

We scan parts of every size with a range of professional 3D scanners — from small precision components to car bodies and full buildings. The result is a precise digital copy of the physical object, ready for product design, reverse engineering and quality control.

High-accuracy capture

Blue-laser and structured-light scanners record detail down to hundredths of a millimetre.

Any size, on site or in-house

From jewellery to vehicles and plant rooms — we bring the scanner to you when the part can't travel.

Clean, usable data

Scans are aligned, cleaned and merged into watertight meshes, not raw point dumps.

Ready for the next step

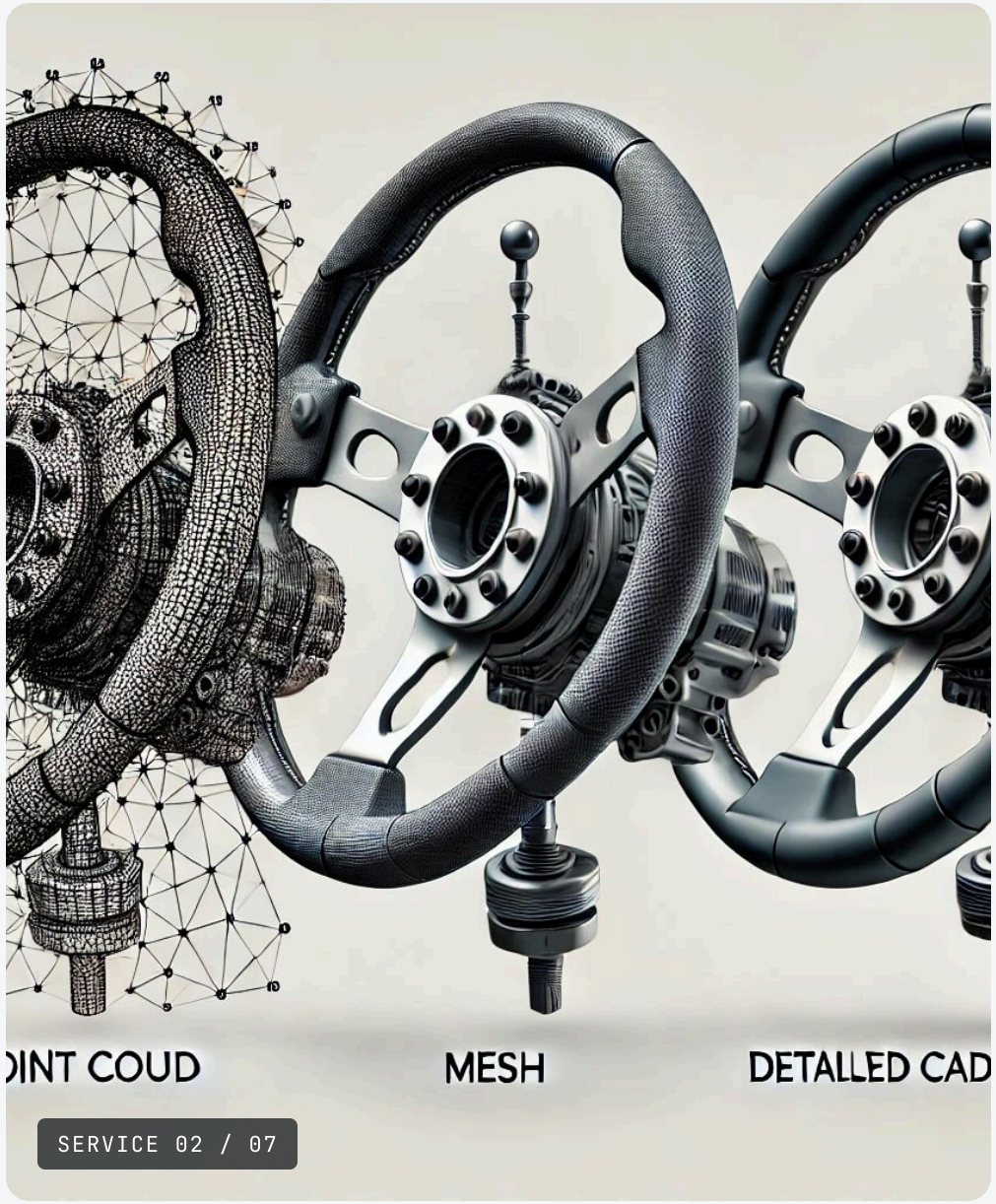
Data is prepared for reverse engineering, inspection, printing or visualisation.

DELIVERABLES

- Point cloud (.PLY, .TXT)
- Mesh (.STL, .OBJ)
- Colour/texture scans
- Scan report

INDUSTRIES

- Aerospace
- Automobile
- Healthcare
- Jewellery
- Education



SERVICE 02 / 07

(02.2) — SERVICES

Reverse Engineering

Turn scan data into fully editable, manufacturable CAD.

We rebuild physical parts as parametric CAD models — for re-manufacturing legacy parts with no drawings, modifying existing designs, or creating moulds and tooling. You choose how faithful the model should be to the scanned part.

Design-intent modelling

A clean CAD model with manufacturing imperfections removed — ideal for re-manufacture from 3D models or 2D drawings.

NURBS surface models

Precise freeform surfaces fitted over scan data where an exact as-is representation matters.

Hybrid modelling

Prismatic features modelled to design intent, freeform areas surfaced to the scan — the best of both.

Drawings on request

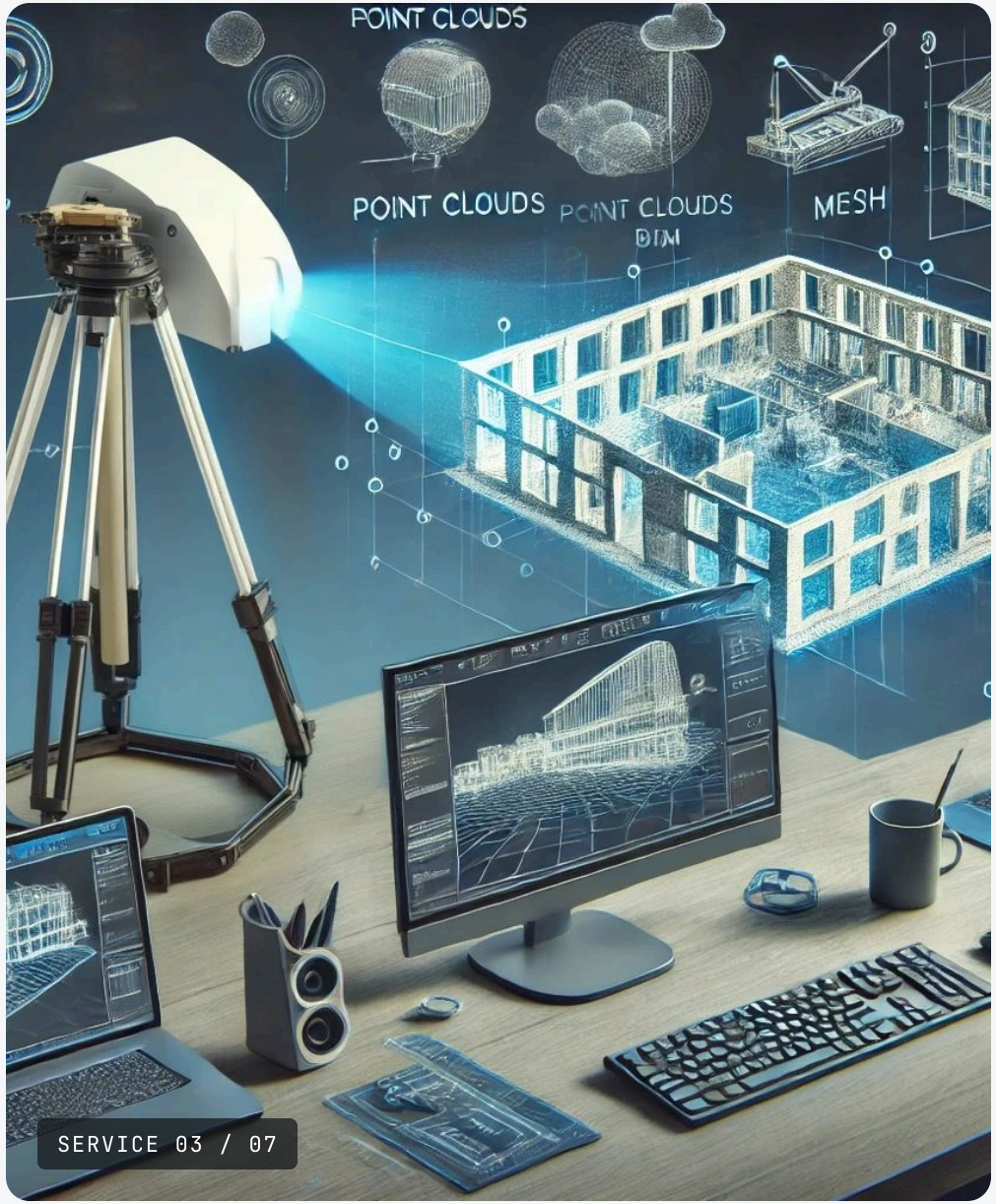
2D manufacturing drawings with GD&T generated from the final model.

DELIVERABLES

- Parametric CAD (.STEP, .IGES, .X_T)
- Native SolidWorks / Creo / NX
- 2D drawings (.PDF, .DWG)
- Deviation report vs. scan

INDUSTRIES

- Automobile
- Aerospace
- Healthcare



(02.3) — SERVICES

Scan to BIM

Existing buildings captured as accurate, intelligent BIM models.

We laser-scan existing structures and convert the point cloud into detailed BIM models. Architects, engineers and contractors get an accurate as-built foundation for renovation, retrofit and facility-management projects — with every wall, slab, column and service documented.

As-built accuracy

Terrestrial laser scanning records the true geometry of the building, not the original drawings.

LOD 200–400 models

Walls, floors, structure and MEP modelled to the level of detail your project needs.

Clash-ready coordination

Models are set up for coordination with new design work in Revit and Navisworks.

Drawings from the model

Floor plans, sections and elevations exported straight from the BIM.

DELIVERABLES

- Registered point cloud (.E57, .RCP)
- Revit model (.RVT)
- IFC export
- 2D plans, sections, elevations

INDUSTRIES

- Construction
- Education
- Healthcare



(02.4) — SERVICES

Class A Surfacing

Flawless, curvature-continuous surfaces for visible parts.

Class A surfaces are the visible skins of a product — car body panels, aircraft fairings, consumer-product housings. We build G2/G3-continuous surfaces with clean patch layouts and verify them with zebra and curvature analysis, so reflections flow without a ripple.

G2 / G3 continuity

Curvature-continuous transitions between every patch for perfect reflections.

Analysis-verified

Zebra stripes, highlight lines and curvature combs checked on every surface.

From scan or sketch

Start from a clay-model scan, a reference surface or concept sketches.

Tooling-ready

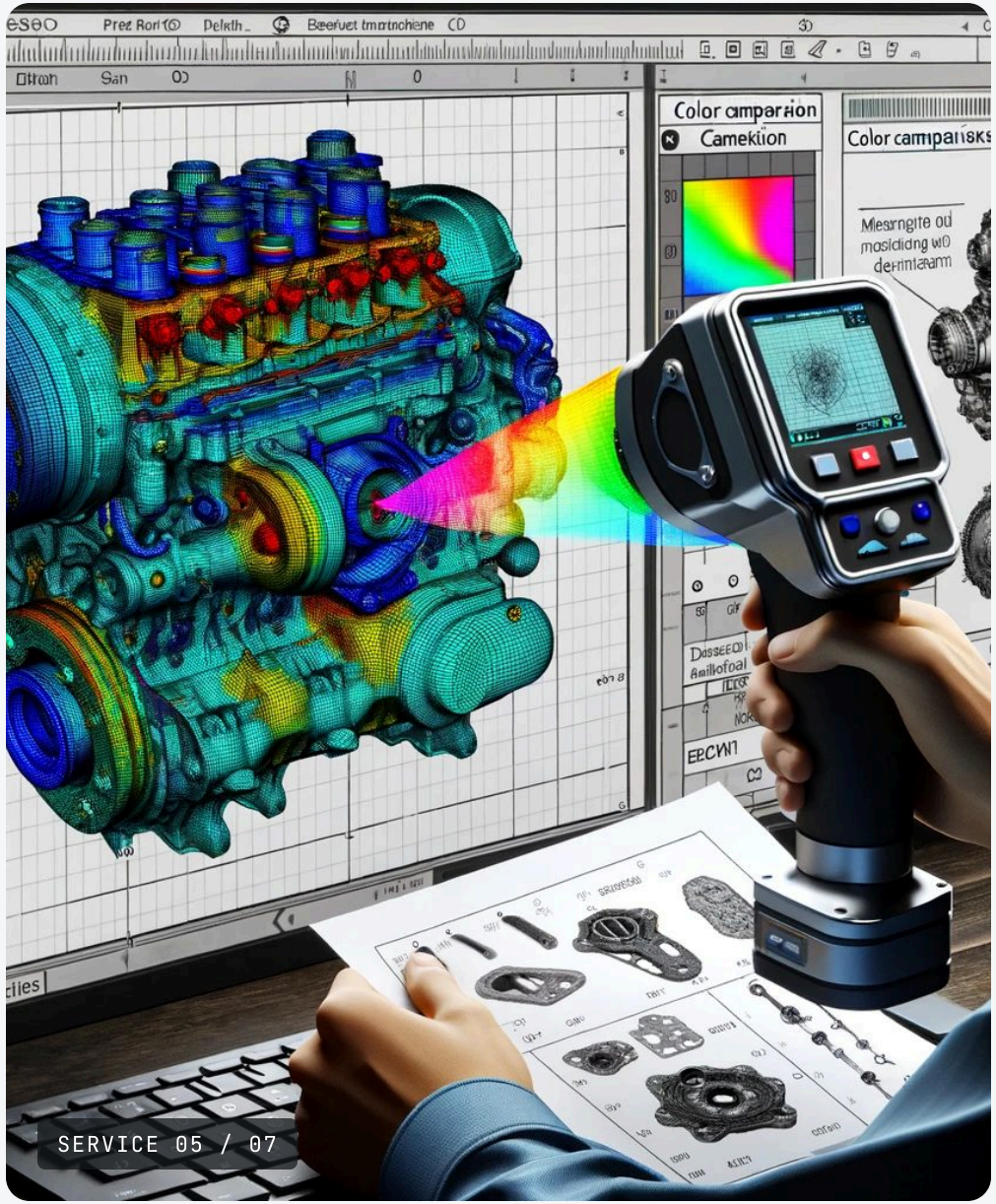
Surfaces handed over with draft, flanges and offsets ready for engineering.

DELIVERABLES

- Class A surfaces (.IGES, .STEP)
- Native Alias / ICEM models
- Surface-analysis reports
- Render-ready model

INDUSTRIES

- Automobile
- Aerospace



(02.5) — SERVICES

Digital Metrology

Compare manufactured parts to their CAD — to the micron.

We inspect manufactured parts by scanning them and comparing the result against the nominal CAD. Colour deviation maps, GD&T checks and first-article reports show exactly where a part is in or out of tolerance — fast, and without contact.

Colour deviation maps

Full-surface comparison against CAD, instantly showing where material is over or under.

GD&T inspection

Flatness, position, profile and more, measured and reported against the drawing.

First-article inspection

Complete FAI reports for new parts, tools and suppliers.

Trend analysis

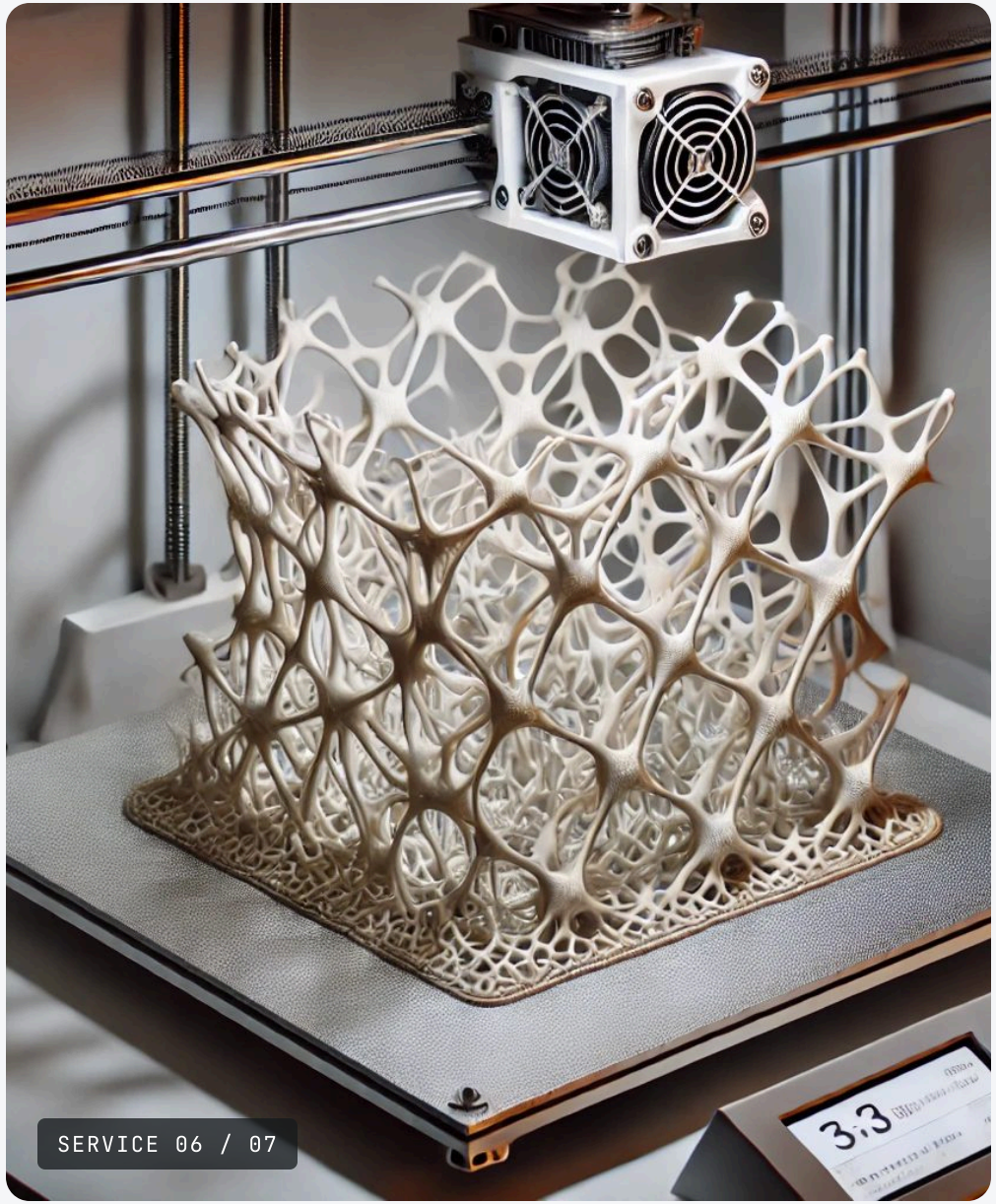
Repeated inspections track tool wear and process drift over time.

DELIVERABLES

- Inspection report (.PDF)
- Colour deviation maps
- GD&T results tables
- Scan data for records

INDUSTRIES

- Aerospace
- Automobile
- Healthcare



(02.6) — SERVICES

3D Printing

Rapid prototypes and functional parts, straight from your model.

We print prototypes, fit-check models, jigs, fixtures and short-run end-use parts directly from CAD or scan data. Our team also prepares and repairs files — including sculpted character models — so they print right the first time.

Rapid prototyping

Physical parts in days to check form, fit and function before tooling.

Functional materials

Engineering plastics and resins chosen for strength, heat or detail.

Print-ready preparation

Mesh repair, hollowing, supports and orientation handled for you.

Character & figure models

Digitally sculpted busts and figures prepared for high-detail resin printing.

DELIVERABLES

- Printed parts
- Print-ready files (.STL, .3MF)
- Finishing / painting on request
- Small production batches

INDUSTRIES

- Education
- Healthcare
- Jewellery
- Automobile



SERVICE 07 / 07

(02.7) — SERVICES

Product Rendering

Photoreal visuals that sell a product before it is ever made.

We turn CAD and design models into photorealistic images and animations for marketing, presentations and design reviews. Materials, lighting and scenes are tuned so your product looks real — and looks its best — before the first unit is built.

Photoreal stills

Studio and lifestyle shots with accurate materials, finishes and lighting.

Animations & turntables

Exploded views, assembly sequences and 360° spins.

Interiors & exhibitions

Spaces, stands and environments visualised before construction.

Jewellery visualisation

Metals and gemstones rendered with true reflections and sparkle.

DELIVERABLES

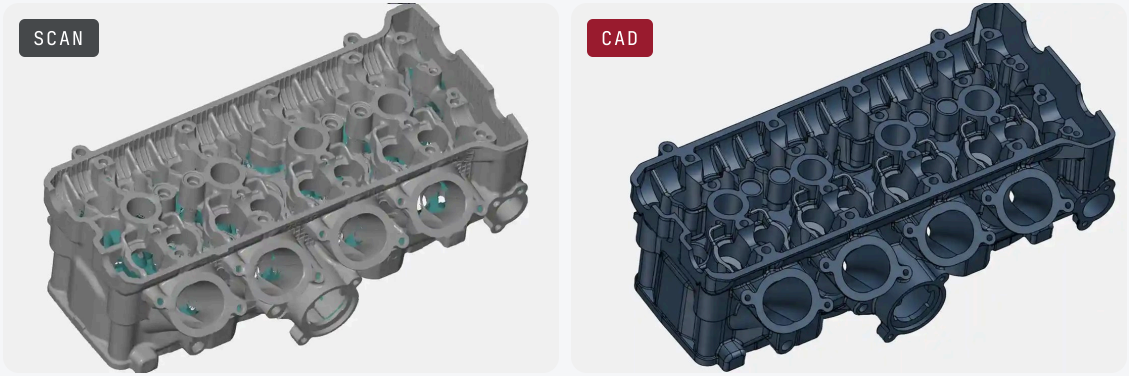
High-res stills (.PNG, .TIFF) · Animations (.MP4) · 360° turntables · Web-ready image sets

INDUSTRIES

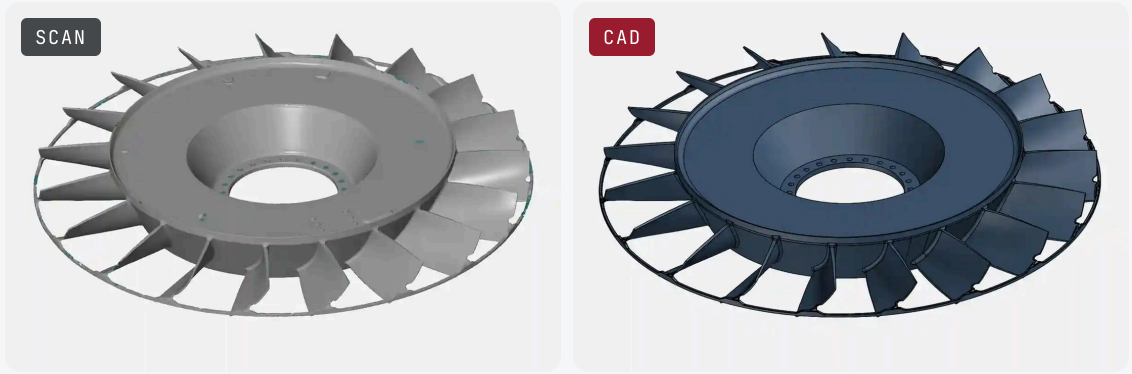
Jewellery · Construction · Automobile

Scan in. CAD out.

Real parts we have captured and rebuilt as editable, manufacturable CAD.



Cylinder Head Engine cylinder head scanned and rebuilt as parametric CAD.



Impeller Bladed impeller captured and modelled with precise freeform blade surfaces.



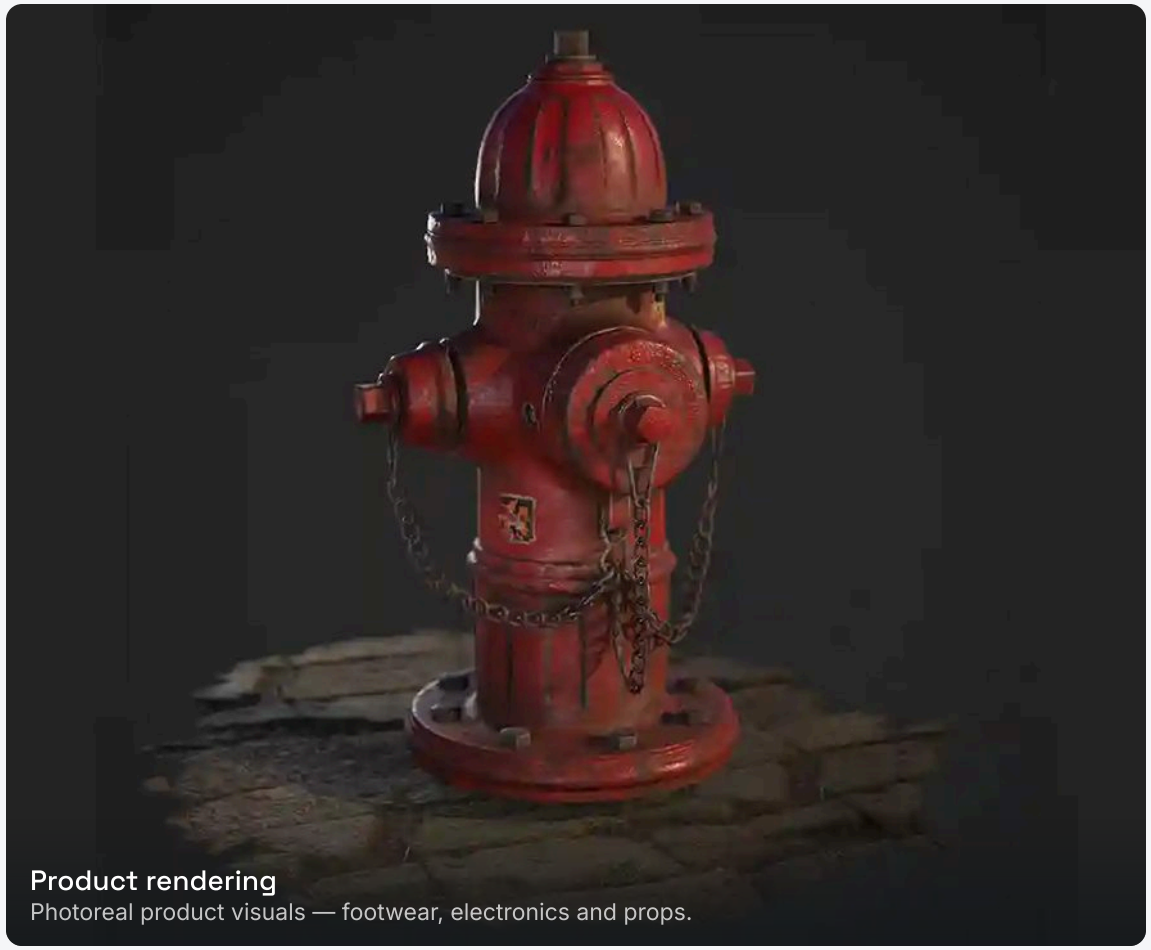
Bike Frame Motorcycle frame digitised and converted into an editable CAD assembly.



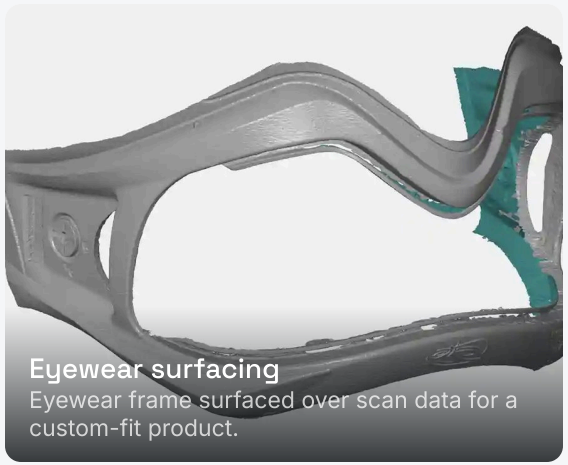
Bumper Vehicle bumper scanned and resurfaced for tooling.

Design & visualisation.

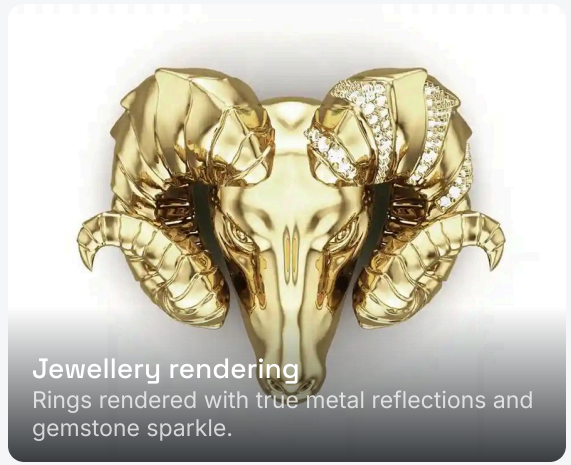
Surfacing, sculpting and photoreal rendering — visuals that sell a product before it is made.



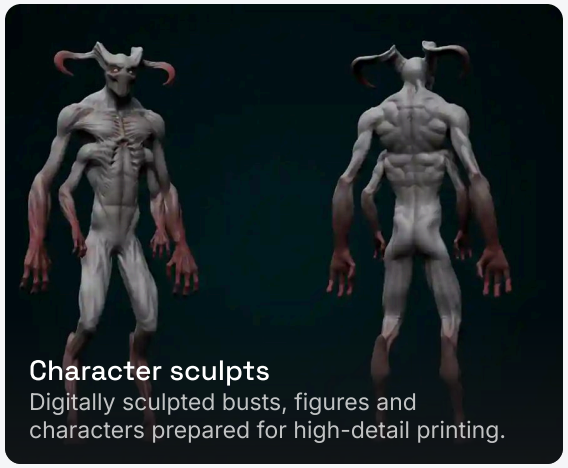
Product rendering
Photoreal product visuals — footwear, electronics and props.



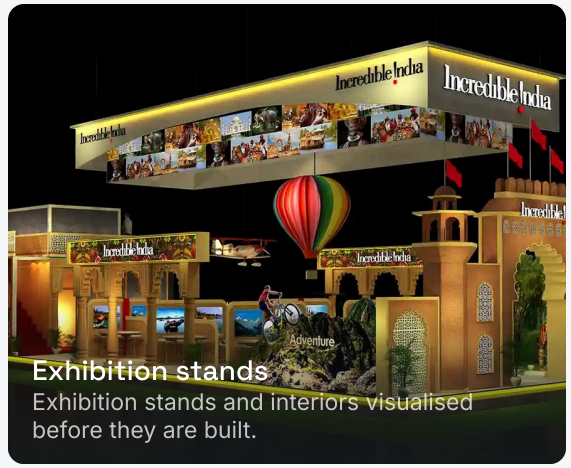
Eyewear surfacing
Eyewear frame surfaced over scan data for a custom-fit product.



Jewellery rendering
Rings rendered with true metal reflections and gemstone sparkle.



Character sculpts
Digitally sculpted busts, figures and characters prepared for high-detail printing.



Exhibition stands
Exhibition stands and interiors visualised before they are built.

Industries we serve.

Six sectors, one standard of precision.



01

Aerospace

Inspection and reverse engineering of high-precision aircraft components.



02

Automobile

Body surfacing, part digitising and fixture design for vehicles.



03

Construction

As-built scanning and BIM models for buildings and sites.



04

Healthcare

Medical devices and custom-fit parts captured and modelled in 3D.



05

Education

Scanning and 3D printing support for labs, research and teaching.



06

Jewellery

Fine-detail scanning and printing for jewellery design and casting.

Professional 3D scanners.

Vertex blue-laser and optical-tracking scanners — now available from VShark, with training and support.



Vertex3D

Handheld blue-laser 3D scanner

0.020 mm precision



Vertex3D Pro

Handheld blue-laser 3D scanner

0.020 mm precision



Vertex Tracker

Optical-tracking 3D scanning system

0.025 mm precision

SPECIFICATION	VERTEX3D	VERTEX3D PRO	VERTEX TRACKER
High-speed scanning	26 crossed blue laser lines	50 crossed blue laser lines	50 crossed blue laser lines
Max. scanning speed	5,400,000 measurements/s	7,100,000 measurements/s	7,100,000 measurements/s
Max. scanning field	650 × 550 mm	650 × 550 mm	1100 × 800 mm
Max. resolution	0.01 mm	0.01 mm	0.01 mm
Volume accuracy	0.015 mm + 0.035 mm/m	0.015 mm + 0.035 mm/m	10.0 m^3: 0.060 mm · 25.0 m^3: 0.083 mm
Recommended sample size	> 10 mm	> 10 mm	10 mm – 8000 mm
Output formats	.STL, .PLY, .ASC, .TXT	.STL, .PLY, .ASC, .TXT	.STL, .PLY, .ASC, .TXT

RECOMMENDED PC Windows 10 or later (64-bit) · NVIDIA GTX/RTX GPU (e.g. RTX 4060 or higher), 8 GB+ VRAM · Intel Core i7-13700H or better · 32 GB RAM or more

(07) — WHAT CLIENTS SAY

Trusted worldwide.

US · UK · GERMANY · CANADA · UAE · SAUDI ARAB

"

VShark Solutions was able to take a relatively poorly scanned object and convert it to a solid model with very accurate dimensions. Their surfacing skills are amazing, and the accuracy is perfect for our needs.

Rachel Labonte
CANADA

"

VSHARK provided the best engineering service we've ever experienced. Their attention to detail and quick responsiveness made working with them a pleasure. We've collaborated with them many times and will continue to do so.

Mace Roper
UNITED STATES

"

Working with VSHARK on a reverse engineering and 3D modelling project was impressive; their professionalism, clear communication, and dedication to quality made the entire process smooth and efficient.

Benedikt Klein
GERMANY

"

Exceptional work and accuracy! I will be using VSHARK skills again; I highly recommend them if you need something converted accurately. No need to look further.

Xander Elon
UNITED KINGDOM

"

VShark Solutions delivered exceptional work beyond my expectations! Their attention to detail and professionalism made the entire process seamless. I highly recommend them for any project.

Ahtisham Muhammad
SAUDI ARABIA

"

Working with VSHARK Engineers was a fantastic experience. Their creative approach and attention to detail perfectly aligned with our brand, and they were always polite and cooperative throughout the project.

Mariam Al-Ghayousi
DUBAI

VSHARK
ENGINEERING SOLUTIONS

(08) — GET IN TOUCH

Let’s start your project.

Send us a photo, drawing or the part itself — we’ll come back with the right approach, timeline and quote.

EMAIL	info@vsharksolutions.com
INDIA	+91 95142 48525
UAE	+971 52 248 6238
WHATSAPP	+91 95142 48525
WEB	vsharksolutions.com